

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 12 34 67 78 98	12 34 67 78 98

Answer:

```
1 #include <stdio.h>
2 void swap(int *a, int *b) {
3     int t = *a; *a = *b; *b = t;
4 }
5 int partition(int arr[], int low, int high) {
6     int pivot = arr[high], i = low - 1;
7     for (int j = low; j < high; j++) {
8         if (arr[j] < pivot) {
9             i++;
10            swap(&arr[i], &arr[j]);
11        }
12    }
13    swap(&arr[i + 1], &arr[high]);
14    return i + 1;
15 }
16 void quickSort(int arr[], int low, int high) {
17     if (low < high) {
18         int pi = partition(arr, low, high);
19         quickSort(arr, low, pi - 1);
20         quickSort(arr, pi + 1, high);
21     }
22 }
23 int main() {
24     int n;
25     if (scanf("%d", &n) != 1) return 0;
26     int arr[n];
27     for (int i = 0; i < n; i++) scanf("%d", &arr[i]);
28     quickSort(arr, 0, n - 1);
29     for (int i = 0; i < n; i++)
30         printf("%d%c", arr[i], (i == n - 1) ? '\n' : ' ');
31     return 0;
32 }
```

	Input	Expected	Got	
✓	5 12 34 67 78 98	12 34 67 78 98	12 34 67 78 98	✓
✓	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✓
✓	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✓

Passed all tests! ✓